



Series · Worksheet 1

Core Pure 2 Chapter 2: Series. Answer all questions. Show your working clearly. Total: 67 marks. Time: 85 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are three sections, one for each topic in Chapter 2. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Unless a question says “from first principles”, you may quote the standard Maclaurin series for e^x , $\sin x$, $\cos x$, $\ln(1+x)$ and $(1+x)^n$. State the range of validity where asked.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Maclaurin series from first principles	21	
B Standard series and the generalised binomial expansion	20	
C Combining series	21	
Roar-level challenge	5	
Total	67	

Section A Maclaurin series from first principles

21 marks

1. Given that $f(x) = e^{3x}$, find $f(0)$, $f'(0)$ and $f''(0)$, and hence find the Maclaurin series of $f(x)$ up to and including the term in x^2 . [2]

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2. Find, from first principles, the Maclaurin series of $\cos 2x$ up to and including the term in x^4 . [3]

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3. Find, from first principles, the Maclaurin series of $\ln(1 + 2x)$ up to and including the term in x^3 . [3]

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4. Show, from first principles, that the Maclaurin series of $\tan x$, up to and including the term in x^3 , is [4]

$$\tan x \approx x + \frac{1}{3}x^3.$$

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5. The function f is defined by $f(x) = e^{\sin x}$.

(a) Show that $f''(x) = (\cos^2 x - \sin x) e^{\sin x}$. [2]

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(b) Find $f'''(x)$. [2]

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(c) Hence find the Maclaurin series of $f(x)$, up to and including the term in x^3 . [3]

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(d) Use your answer to part (c) to find an approximate value for $\int_0^{0.2} e^{\sin x} dx$, giving your answer to 3 decimal places. [2]

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(Total for Question 5 is 9 marks)

**Section B Standard series and the generalised binomial expansion**

20 marks

6. Write down the first four terms of the series expansion of e^{-2x} , in ascending powers of x . [2]

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7. Find the first three non-zero terms of the series expansion of $\ln(1 - 3x)$, in ascending powers of x , and state the range of values of x for which the expansion is valid. [3]

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8. Find the series expansion of $\sqrt{1 + 4x}$, in ascending powers of x , up to and including the term in x^3 . State the range of values of x for which the expansion is valid. [3]

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9. Find the series expansion of $\frac{1}{\sqrt{4-x}}$, in ascending powers of x , up to and including the term in x^3 . State the range of values of x for which the expansion is valid. [4]

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10. (a) Find the series expansion of $\sqrt{4+x}$, in ascending powers of x , up to and including the term in x^3 , simplifying each coefficient. State the range of values of x for which the expansion is valid. [4]

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- (b) By substituting $x = 0.04$ into your expansion, find an approximation for $\sqrt{101}$, giving your answer to 6 decimal places. [3]

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- (c) Explain why the substitution $x = 97$ could not be used to approximate $\sqrt{101}$. [1]

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(Total for Question 10 is 8 marks)



Section C Combining series

21 marks

11. Using the series expansion of e^x , find the first three non-zero terms of the series expansion of e^{x^2} . [2]

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12. Find the series expansion of $e^x \sin x$, in ascending powers of x , up to and including the term in x^3 . [3]

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13. Show that, for $-1 < x < 1$, [3]

$$\ln\left(\frac{1+x}{1-x}\right) = 2x + \frac{2}{3}x^3 + \frac{2}{5}x^5 + \dots$$

and explain why the expansion is valid only for $-1 < x < 1$.

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14. By writing $1+x+x^2 = \frac{1-x^3}{1-x}$, find the series expansion of $\ln(1+x+x^2)$, in ascending powers of x , up to and including the term in x^4 . [4]

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15. (a) Using the series expansions of $\cos x$ and $\ln(1+x)$, show that [5]

$$\ln(\cos x) \approx -\frac{1}{2}x^2 - \frac{1}{12}x^4.$$

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- (b) Hence write down the series expansion of $\ln(\sec x)$ up to and including the term in x^4 . [1]

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- (c) Use your answer to part (a) to find an approximate value for $\int_0^{0.5} \ln(\cos x) dx$, giving your answer to 4 decimal places. [3]

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(Total for Question 15 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

16. Given that $f(x) = \frac{e^{2x}}{1+x}$, find the series expansion of $f(x)$, in ascending powers of x , up to and including the term in x^3 , and state the range of values of x for which it is valid. [5]
- Hence find $\lim_{x \rightarrow 0} \frac{f(x) - 1 - x}{x^2}$.

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END OF WORKSHEET

Total: 67 marks